INCH-POUND
MIL-DTL-24643/61B
27 November 2023
SUPERSEDING
MIL-DTL-24643/61A
28 January 2010

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +176 DEGREES FAHRENHEIT (+80 DEGREES CELSIUS), 150 VOLTS, TYPES LSC5P, LSC5POS, AND LSC5POSR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable can be manufactured in three variations: LSC5P (24 AWG, Unshielded Patch Cable), LSC5POS (24 AWG, Metallic Foil and Braid Patch Cable), and LSC5POSR (26 AWG, Metallic Foil and Braid Patch Cable).

Construction (Non-Watertight)

First - Copper conductor, tin-coated, 7 strands, in accordance with ASTM B286, 24 AWG or 26 AWG (see table I for applicable size).

Second - Polyolefin insulation. Wall thickness as necessary in order for the completed cable to comply with the electrical requirements specified herein. Maximum insulation diameter shall not exceed 0.046 inch.

Third - Two insulated conductors, twisted together with an appropriate lay length for each pair such that the completed cable complies with the electrical requirements specified herein. Wire identification codes shall be in compliance with ANSI/TIA 568.2 as follows:

Pair 1 – white/blue and blue
Pair 2 – white/orange and orange
Pair 3 – white/green and green
Pair 4 – white/brown and brown

Fourth - Four pairs shall be cabled together with a lay as required to comply with the electrical requirements specified herein. Cabling sequence shall be consecutive. Fillers or other means may be used as necessary to form a firm, well-rounded cable that complies with the electrical requirements specified herein.

Fifth - An optional binder.

Construction for Variant Type LSC5P

Sixth - Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.040 inch. See table I for cable overall diameter.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

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Construction for Variant Types LSC5POS and LSC5POSR

Sixth - Metallic foil/polyester tape (metallic foil side facing out and in contact with the braid). Tape may be longitudinally or helically applied, with a minimum overlap of 10 percent.

Seventh - Braided shield of 36 AWG or 34 AWG, tin-coated copper.

Eighth - An optional binder.

Ninth - Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.040 inch. See table I for cable overall diameter.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

TABLE I. Details.

Military part no. M24643/61	AWG	Type and size	Overall diameter (inch)	
			min.	max.
-01UN	24	LSC5P-4	0.230	0.325
-02UO	24	LSC5POS-4	0.290	0.370
-03UO	26	LSC5POSR-4	0.230	0.325

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/100 meters at 68 °F [20 °C], max.)	
26 AWG	15.20
24 AWG	9.38
Conductor resistance unbalance (percent, max.)	5
Voltage withstand	
Conductor to conductor	1.5 kVAC or 2.5 kVDC
Conductor to overall shield (types LSC5POS and LSC5POSR only)	1.5 kVAC or 2.5 kVDC
Insulation resistance (megohms/100 meters, max.)	Not applicable
Conductor continuity	No failure
Shield continuity (types LSC5POS and LSC5POSR only)	No failure
Jacket flaws (types LSC5POS and LSC5POSR only)	No failure

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Group A:

Visual and dimensional	No failure
Capacitance	
Mutual capacitance (nominal at 1 kilohertz [kHz]) (nanofarad [nF]/100 meters)	5.6
Pair-to-ground capacitance unbalance (max. at 1 kHz) (picofarad [pF]/100 meters)	330
Input impedance (at 1 megahertz [MHz] to 100 MHz) (ohms ±15%)	100
Swept electrical parameters	See table II

TABLE II. Swept electrical parameters.

Frequency (MHz)	1.0	4.0	10	16	31.25	62.5	100
Return loss dB/100 m (min.)	20.0	23.0	25.0	25.0	23.6	21.5	20.1
Insertion loss dB/100 m, 24 AWG (max.)	2.4	4.9	7.8	9.8	14.0	20.4	26.4
Insertion loss dB/100 m, 26 AWG (max.)	3.0	6.2	9.8	12.3	17.6	25.5	33.0
NEXT dB/100 m (min.)	65.3	56.3	50.3	47.2	42.9	38.4	35.3
ELFEXT dB/100 m (min.)	63.8	51.8	43.8	39.7	33.9	27.9	23.8
PS NEXT dB/100 m (min.)	62.3	53.3	47.3	44.2	39.9	35.4	32.3
PS ELFEXT dB/100 m (min.)	60.8	48.8	40.8	36.7	30.9	24.9	20.8
Propagation delay ns/100 m (max.)	570	552	545	543	540	539	538
Delay skew ns/100 m (max.)	45	45	45	45	45	45	45

NOTE:

1. Values are for reference only. Actual values shall be determined utilizing the formulas in ANSI/TIA 568.2.

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Group B:	
Cold bending (cable)	No failure
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Thermoset proof test (percent, max.)	
Insulation	Not applicable
Jacket (when tested at 392 °F [200 °C])	50
Group C:	
Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (jacket) (cycles, min.)	125
Shrinkage	No failure
Heat distortion (percent of unaged, max.)	30
Surface transfer impedance (types LSC5POS and LSC5POSR only)	
Milliohms per meter (max.)	700
Electromagnetic pulse (EMP) response (dB, min.)	60
Group D:	
Flame propagation (cable)	No failure

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Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable) (203±3.6 °F [95±2 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	5
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	45
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	Not applicable
Stress resistance	Not applicable
Temperature humidity	Not applicable

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
All types and sizes	500

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CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-077)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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